

JEE MAIN REVISION MASTER: PARABOLA SYLLABUS REDUCED

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 52 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Standard Parabola, Focus, Directrix & Tangents	49 PYQs	Parabola
Moderate	Normals to Parabola & Co-Normal Points	4 PYQs	Normals

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Standard Parabola, Focus, Directrix & Tangents	$y^2 = 4ax \implies \text{Focus} (a, 0), \text{Directrix: } x = -a, \text{LR} = 4a, \text{Tangent: } y = mx + \frac{a}{m}, \text{Parametric: } (at^2, 2at)$
Normals to Parabola & Co-Normal Points	$\text{Normal: } y = mx - 2am - am^3, \text{Parametric: } y + tx = 2at + at^3, \sum m_i = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Standard Parabola, Focus, Directrix & Tangents (Very High Yield — 49 PYQs)

Core Principle: Focal properties of $y^2 = 4ax$, slope form of tangent $y = mx + a/m$, and parametric chords.

■ **Caution / JEE Pitfall:** For $x^2 = 4ay$, the tangent in slope form is $y = mx - am^2$, NOT $y = mx + a/m$!

■ **Speed Trick:** The tangents at the extremities of any focal chord intersect at right angles (90°) strictly on the DIRECTRIX.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let O be the vertex of the parabola $y^2=4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of $th...$ " [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "Let the parabola $y=x^2+px+q$ passing through the point $(1,-1)$ be such that the distance between its vertex and the x -axis is minimum..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Let O be the vertex of the parabola $x^2=4y$ and Q be any point on it. Let the locus of the point P , which divides the line segment OQ in..." [Ans: N/A]

3.2 Normals to Parabola & Co-Normal Points (Moderate Yield — 4 PYQs)

Core Principle: Cubic normal equation $am^3 + (2a - x)m + y = 0$, maximum of three normals from external point, and co-normal properties.

■ **Caution / JEE Pitfall:** From an external point (h, k) , three normals can be drawn to $y^2 = 4ax$ if and only if $h > 2a$.

■ **Speed Trick:** The algebraic sum of the slopes of three co-normal points is always ZERO: $m_1 + m_2 + m_3 = 0$.

Representative Source PYQs:

- **[2024 Morning Shift] Q#1 (MCQ):** "If the shortest distance of the parabola $y^2=4x$ from the centre of the circle $x^2+y^2-4x-16y+64=0$ is d , then d^2 is equal to..." **[Ans: N/A]**
- **[2023 Evening Shift] Q#3 (MCQ):** "The parabolas : $ax^2+2bx+cy=0$ and $dx^2+2ex+fy=0$ intersect on the line $y=1$. If a, b, c, d, e, f are positive real numbers and $a, b, c, \dots$ " **[Ans: N/A]**
- **[2023 Morning Shift] Q#3 (MCQ):** "The distance of the point $(6, -2\sqrt{2})$ from the common tangent $y=mx+c, m>0$, of the curves $x=2y^2$ and $x=1+y^2$ is ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Let O be the vertex of the parabola $y^2=4x$ and its chords OP and OQ are perpendicular to each other. If the locus of the mid-point of the line segment PQ is a conic C , then the length of its latus rectum is :

- (A) 1
- (B) 2
- (C) 4
- (D) 8

Correct Answer: None

Solution Analysis: $t_1(1)t_2(2)=-4$ $2k=t_1(1)^2+t_2(2)^2$ $k=2(t_1(1)+t_2(2))$ $2k=k^2-2t_1(1)t_2(2)$ $2k=k^2+8$ $k^2-2k-8=0$ $k=4$ or $k=-2$
L.R.=2

Benchmark Problem 2: JEE Main 2023 (Online) 30th January Evening Shift (Q#3)

The parabolas : $ax^2+2bx+cy=0$ and $dx^2+2ex+fy=0$ intersect on the line $y=1$. If a, b, c, d, e, f are positive real numbers and a, b, c are in G.P., then :

- (A) $(d/a), (e/b), (f/c)$ are in A.P.
- (B) $(d/a), (e/b), (f/c)$ are in G.P.
- (C) d, e, f are in A.P.
- (D) d, e, f are in G.P.

Correct Answer: None

Solution Analysis: Let point of intersection be $(\alpha, 1)$ $a\alpha^2+2b\alpha+c=0$ (i) and $d\alpha^2+2e\alpha+f=0$ (ii) $\Rightarrow a\alpha^2+2\sqrt{ac}\alpha+c=0$ ($b^2=ac$) $(\sqrt{a}\alpha+\sqrt{c})^2=0$ $\alpha=-\sqrt{(c/a)}$ Put the value of α in (ii), $d(c/a)-2e\sqrt{(c/a)}+f=0$ $(d/a)-\frac{2e}{\sqrt{a}}\sqrt{(c/a)}+f/c=0$ $(d/a)+(f/c)=2(e/b)$ $(d/a), (e/b), (f/c)$ are in A.P.